

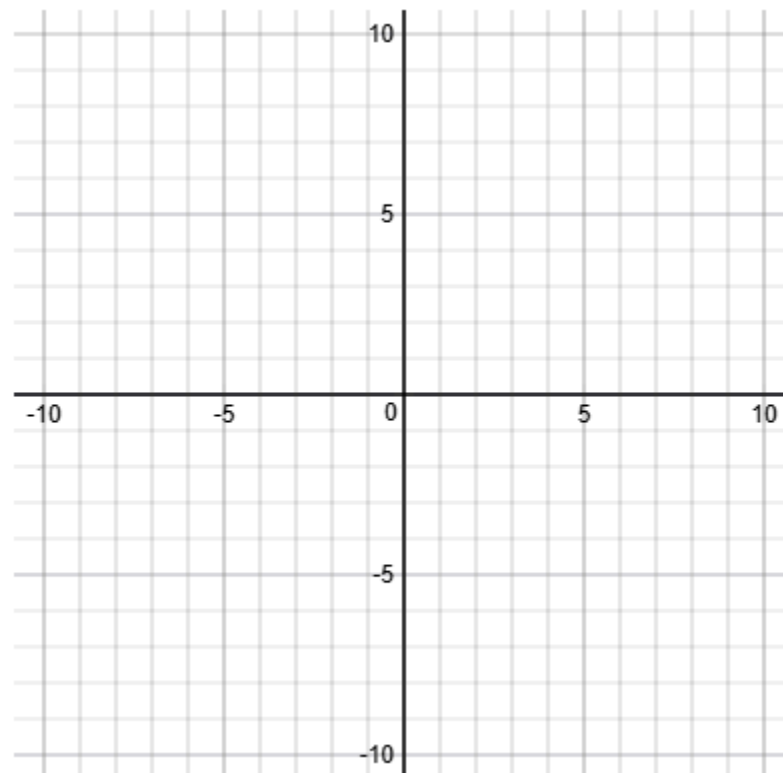
Calculus I Lesson 42

1) $f(x) = \frac{3}{4}x + 6$

a) $f(x) = \frac{3}{4}x + 6$ a one-to-one function?

b) Find $f^{-1}(x)$.

c) Graph $f(x)$, $f^{-1}(x)$, and the line $y = x$.

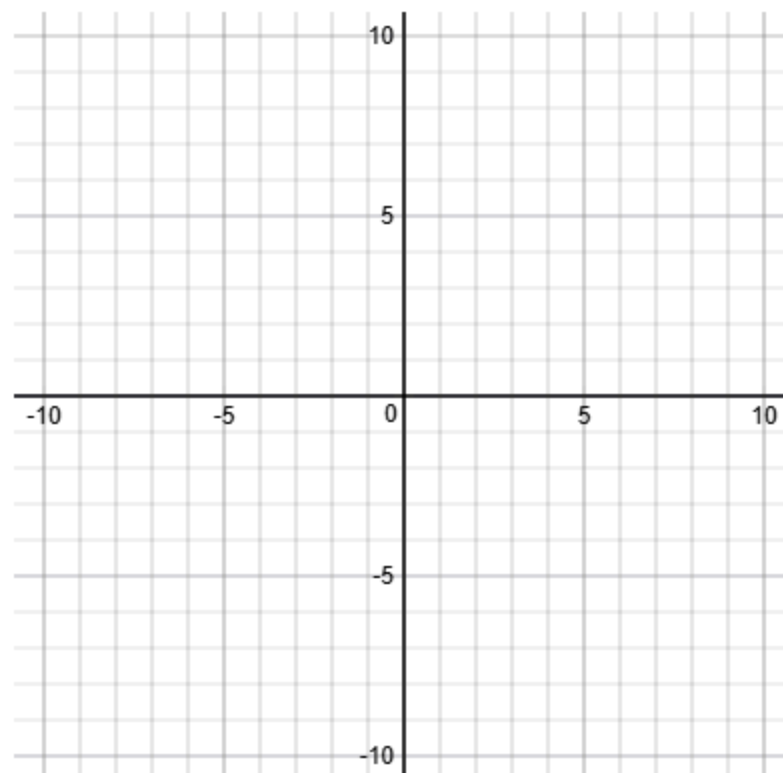


2) $f(x) = 2x - 3$

a) $f(x) = 2x - 3$ a one-to-one function?

b) Find $f^{-1}(x)$.

c) Graph $f(x)$, $f^{-1}(x)$, and the line $y = x$.

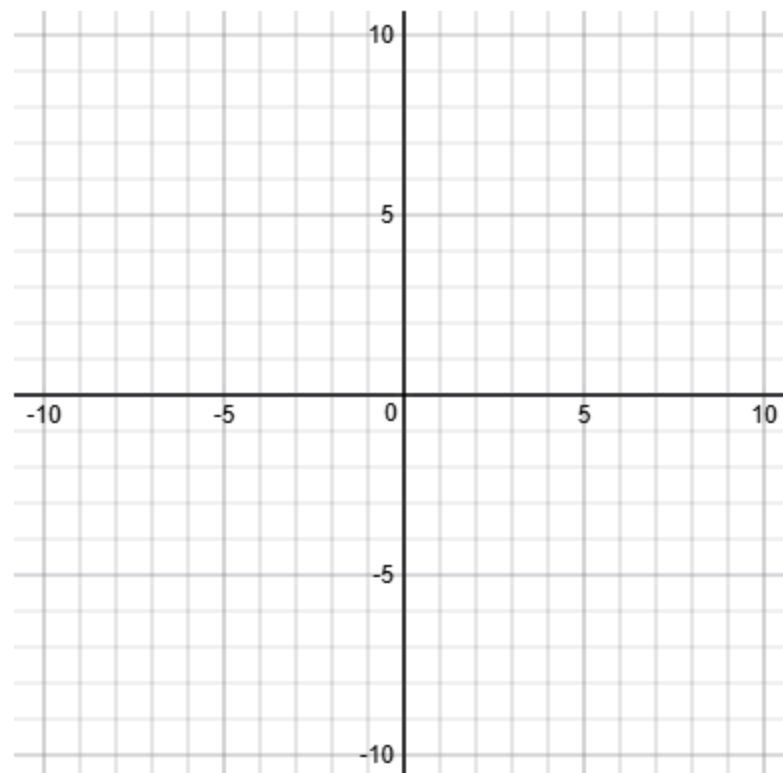


3) $f(x) = x^4$

a) Is $f(x)$ a one-to-one function?

b) Find $f^{-1}(x)$.

c) Graph $f(x)$, $f^{-1}(x)$, and the line $y = x$.

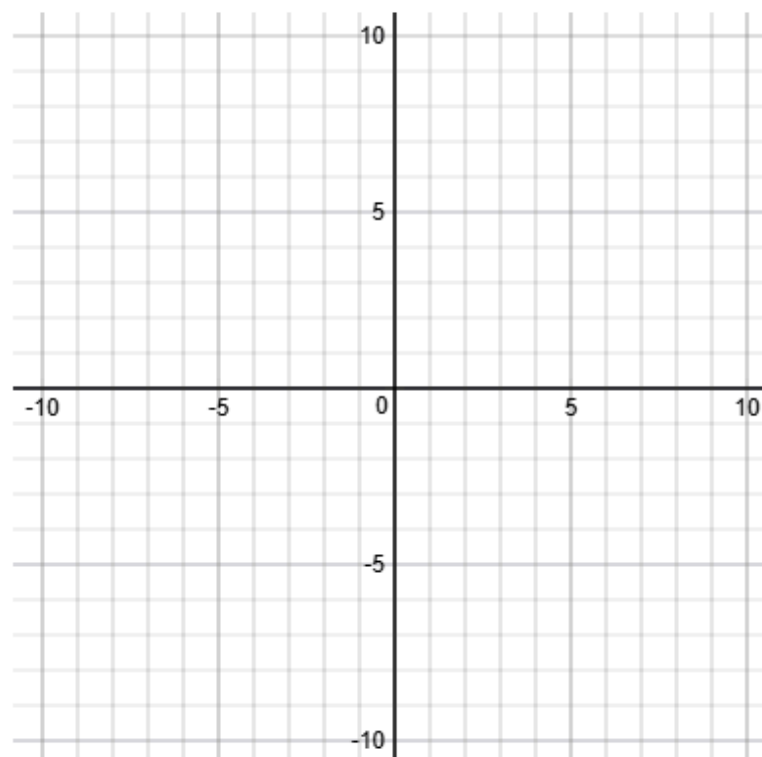


4) $f(x) = \sqrt{25 - x^2}$, $0 \leq x \leq 5$

a) Is $f(x)$ a one-to-one function?

b) Find the inverse function $f^{-1}(x)$.

c) Graph $f(x) = \sqrt{25 - x^2}$, $f^{-1}(x)$, and the line $y = x$.

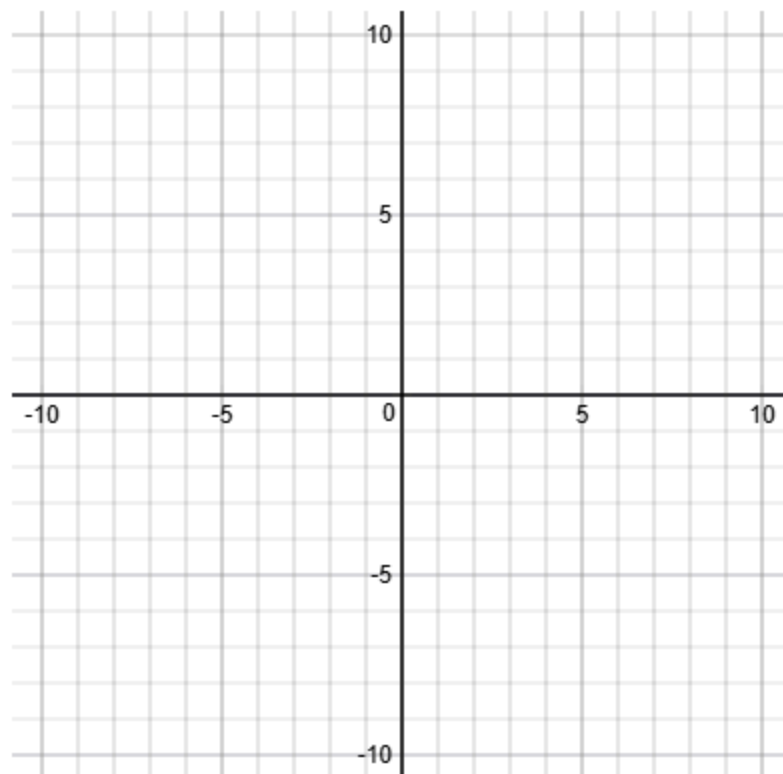


5) $f(x) = \sqrt[3]{x-4}$

a) Is $f(x) = \sqrt[3]{x-4}$ a one-to-one function?

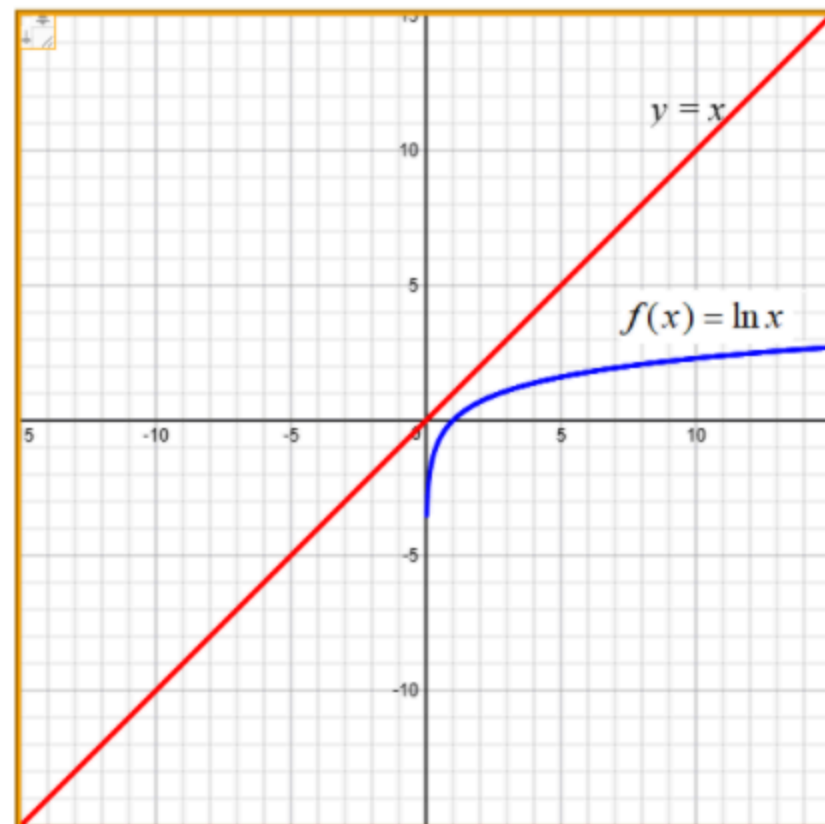
b) Find $f^{-1}(x)$.

c) Graph $f(x)$, $f^{-1}(x)$, and the line $y = x$.



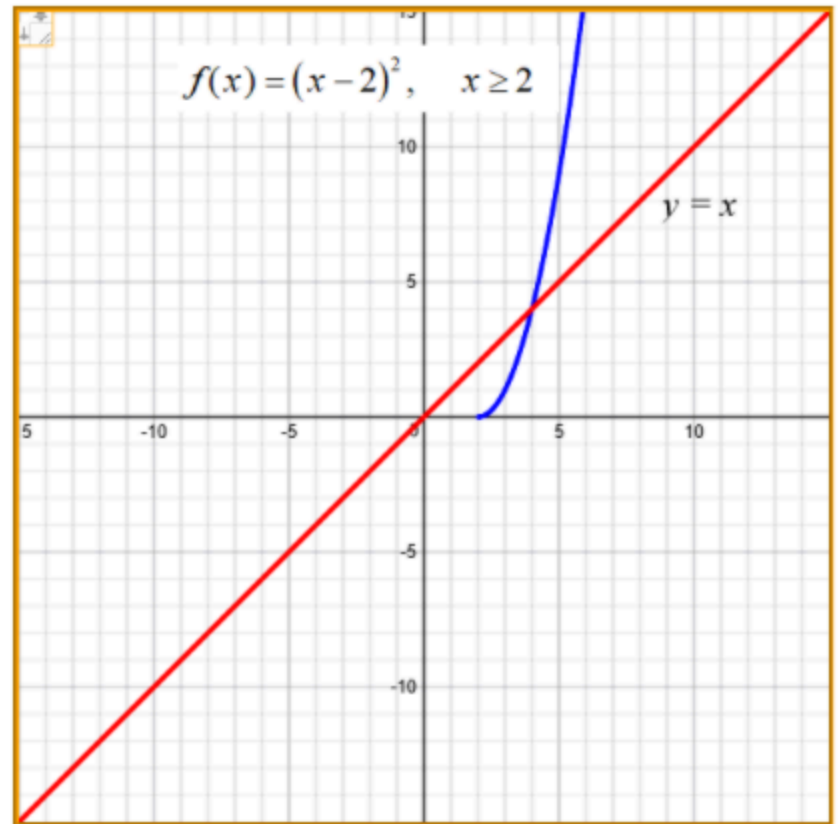
6) Let $f(x) = \ln x$

Draw the image of the inverse function of $f(x)$.



7) $f(x) = (x - 2)^2$, $x \geq 2$

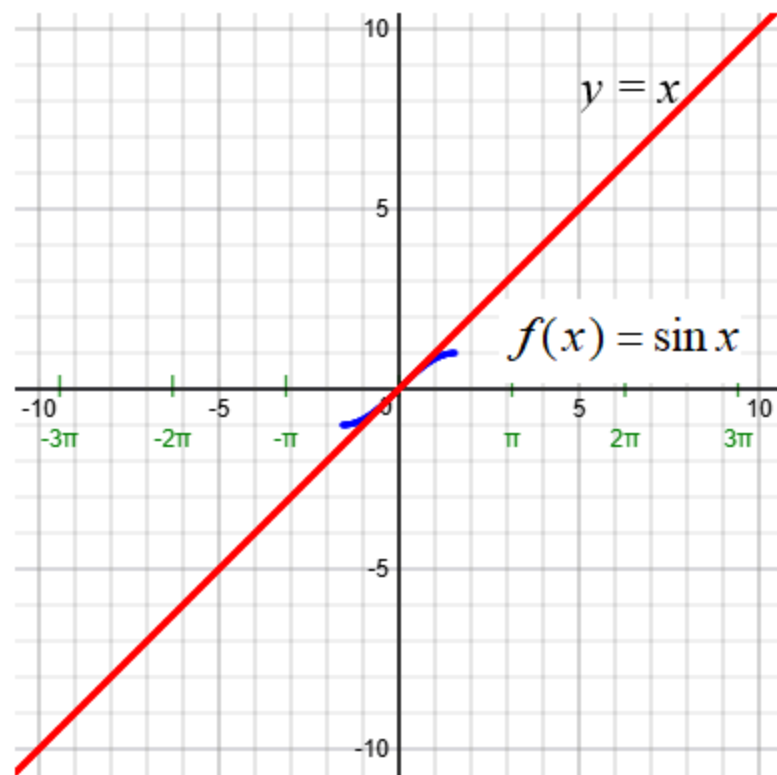
Draw the image of the inverse function of $f(x)$.



8) $f(x) = \sin x, \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

The inverse function is $f^{-1}(x) = \sin^{-1} x$,

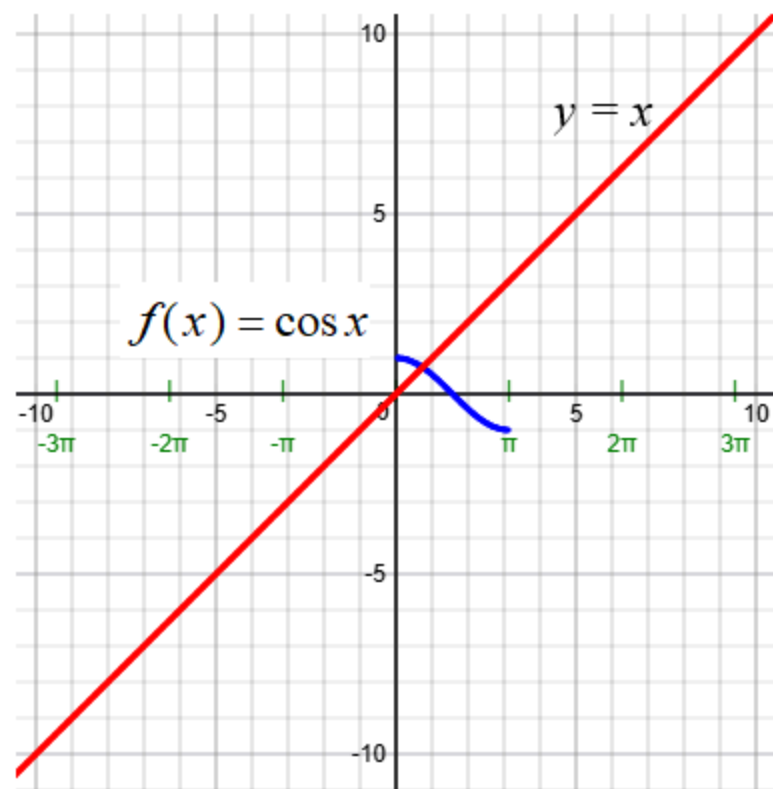
Draw the image of the inverse function of $f(x)$.



9) $f(x) = \cos x, \quad 0 \leq x \leq \pi$

The inverse function is $f^{-1}(x) = \cos^{-1} x$,

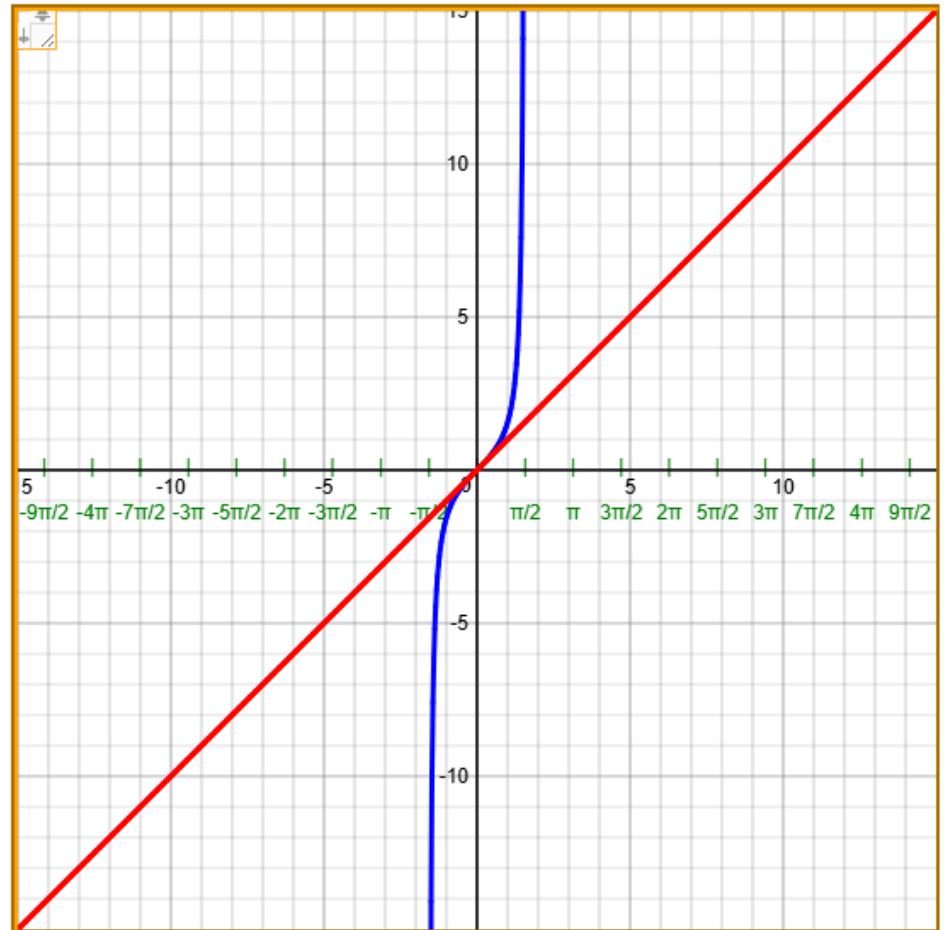
Draw the image of the inverse function of $f(x)$



10) $f(x) = \tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$

The inverse function is $f^{-1}(x) = \tan^{-1} x$,

Draw the image of the inverse function of $f(x)$



11) $f(x) = \tan x$ and $g(x) = \tan^{-1} x$

a) $\lim_{x \rightarrow \infty} \tan^{-1} x = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow -\infty} \tan^{-1} x = \underline{\hspace{2cm}} ?$

